

Sri Lanka Institute of Information Technology

Faculty of Computing

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IT2130 - Operating Systems and System Administration

Year 02 and Semester 02

Lecture 06

Protection and Security of the Operating System

Outline

- Goals of Protection
- The Security Problem
- Reasons for taking Security Measures
- Access Matrix
- Program Threats
- System and Network Threats

Protection and Security

- Operating System uses two sets of techniques to counter threats to information namely;
 - **Protection** - It involves guarding a user's data and programs against inference by other authorized users of the system.
 - **Security** - It involves guarding of a user's data and programs against interference by external entities. Eg: Unauthorized persons.

Facets to Protection of Information

- There are two facets to protection of information
 - **Secrecy**: implies that only authorized users should be able to access information.
 - **Privacy**: implies that information should be used only for the purpose(s) for which it is intended and shared.
- Operating System focuses on guaranteeing secrecy of information and leaves the issues of privacy to the users and their processes.

Security and Protection: Policies and Mechanisms

Security

- **Policy:** Specify whether a person can become a user of the system.
- **Mechanism:**
 - Add or delete users
 - Verify whether a person is an authorized user.

Security and Protection: Policies and Mechanisms

Protection

- **Policy:** Specify whether a user can access a specific file.
- **Mechanism:**
 - Set or change protection information of a file.
 - Check whether a file can be accessed by a user.

Why is Security a Problem

- Modern operating systems face security challenges because:
 - Multiple users share the same system
 - Programs may not be trustworthy
 - Systems are connected to networks (Internet)
 - Attackers continuously evolve new techniques

Reasons for taking Security Measures

- To prevent loss of data
- To prevent corruption of data
- To prevent compromise of data
- To prevent theft of data

Authentication

- The goal of authentication is the reasonable assurance that anyone who attempts to access a system or a network is a legitimate user.
- There are 3 Mechanisms:
 - ① Password
 - ② Physical token or an artifact
 - ③ Biometric measure

Core Security Principles

- Operating System security is built around 3 objectives known as CIA;
 - Confidentiality
 - Integrity
 - Availability



Core Security Principles

- **Confidentiality** - The information is not disclosed to unauthorized individuals, entities and process.

Eg: If someone sees my computer password while I am logging in, then my password has been compromised, and confidentiality has been breached.



Core Security Principles

- **Integrity** - This ensures that data and system resources are not modified without authorization.
Eg: Only authorized users can modify system files
- **Availability** - Availability ensures that system resources are accessible when needed.
Eg: CPU time is fairly shared among processes.



Security Models

- Security models can be discretionary or mandatory.
 - Discretionary: Holders of right can be allowed to transfer them at their discretion.
 - Mandatory: Only designated roles are allowed to grant rights.

Security Policy Vs. Security Model

- Security Policy: Outlines several high-level points:
 - How the data is accessed
 - The amount of security required and what are the steps when these requirements are not met.
- Security Model: The mechanism to support the security policy. This involves in the design of the security system.

Access Matrix Model

- This consists of three principal components:
 - A set of **passive objects** (files, terminals, devices and other entities)
 - A set of **active subjects**, which may manipulate the objects
 - A set of **rules** governing the manipulation of objects by subjects.

- The access matrix is a rectangular array with one row per subject and one column per object.

Objects →	File 1	File 2	File 3
Subjects			
↓			
User 1	r,w	r	r,w,x
User 2	r	r	r,w,x
User 3	r,w,x	r,w	r,w,x

- Each cell defines what operations a subject is allowed to perform on an object.

Program Threats

- Program threats occur when a malicious program exploits system vulnerabilities to gain unauthorized access or disrupt system operation.
- There are two types of threats:
 - Direct: This is any direct attack on your specific systems, whether from outside hackers or from disgruntled insiders.
 - Indirect: This is general random attack, most commonly computer viruses, computer worms and Trojan horses.

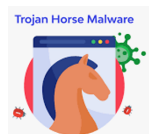
Common Threats

1. Trojan Horse

- A malicious program disguised as a legitimate one
- Performs hidden harmful actions
- Eg: A fake “calculator” that steals passwords

2. Trapdoor (Backdoor)

- A hidden entry point in a program
- Bypasses normal authentication
- Eg: Developer leaves a secret login for testing



Common Threats

3. Logic Bomb

- Malicious code that activates when a specific condition is met
- Eg: Code deletes files on a specific date



4. Virus

- Attaches itself to a legitimate program
- Replicates when the program runs
- There can be file corruptions, system slowdown and data losses



Common Threats

5. Worm

- Self-replicating program
- Spreads without user interaction, often via networks
- These are different from Virus because worms do not need a host program



OS-Level Defenses

- Code signing
- Antivirus software
- Sandboxing
- Least privilege execution

System and Network Threats

- Threats that target the entire system or network infrastructure rather than individual programs.
- There are different system threats:

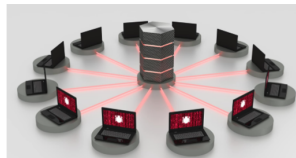
1. Denial of Service (DoS)

- Makes system resources unavailable
- Overloads CPU, memory, or network
- Eg: Flooding a server with requests

System and Network Threats

2. Distributed Denial of Service (DDoS)

- DoS attack from multiple machines (botnet)



3. Spoofing

- Attacker pretends to be a legitimate user or system
- There are two types: IP spoofing and Email spoofing

4. Port Scanning

- Scanning open ports to find vulnerabilities

System and Network Threats

5. Man-in-the-Middle (MITM)

- Attacker intercepts communication between two parties
- Eg: Stealing passwords on public Wi-Fi
- Network-Based Malware
 - Network worms
 - Ransomware
 - Spyware

OS and Network Defenses

- Firewalls
- Intrusion Detection Systems (IDS)
- Secure protocols (HTTPS, SSH)
- Regular patching and updates

End of Lecture 6

Weekly plan

- **Practical:** Pipe implementation in C
- **Workshop:**
 - Summary – Protection and Security of the OS
 - Discussion of Tutorial 6

